NU MOA

Week 3: GEITE01X
Living in the IT Era

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Digital Technology and Social Change

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Agenda for Today:

1. Opening Prayer
2. Greetings and Checking of Attendance
3. Review of the Past Lesson (short recitation)
4. New Lesson: Week 3 and 4 (class discussion and interaction)
5. Q and A
6. Closing Prayer

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LEARNING OUTCOMES

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At the end of this lesson, you should be able to:

1. Understand the role of digital technology in shaping social dynamics, communication, and interaction patterns, and recognize its influence on various aspects of society.
2. Articulate basic, but fundamental definitions of complex issues and dynamics that humans encounter every day, such as technology, social progress, development, and digitalization
3. Critically examine how digital technology intersects with social challenges, such as privacy concerns, misinformation, cyberbullying, and digital divide, and propose strategies to address these issues.

BEFORE THE 0'S AND 1'S

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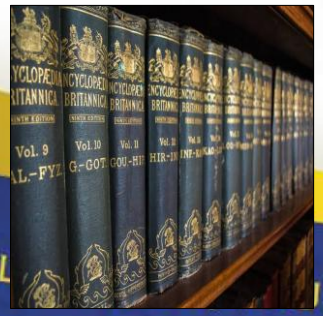
	BEFORE	AFTER
Communication:	Primarily face-to-face, handwritten letters, telegrams, and landline telephones.	Instant messaging, emails, video calls, and social media platforms enable real-time global interaction.
Information Access:	Relied on printed materials, such as newspapers, books, and encyclopedias.	Knowledge is instantly available online through search engines, e-books, online journals, and digital databases.
Media Consumption:	News and entertainment came from newspapers, radio, and television.	People use streaming services, online news portals, podcasts, and social media feeds.
Entertainment:	Included traditional games, theaters, outdoor activities, and physical media like vinyl records and VHS tapes.	Expanded to digital gaming, virtual reality, online streaming platforms (e.g., Netflix, Spotify, YouTube), and social networking.

BEFORE THE 0'S AND 1'S

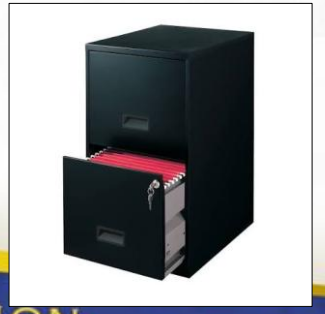
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	BEFORE	AFTER
Work:	Involved manual processes, typewriters, filing cabinets, and paper-based documentation.	Shifted to computers, cloud systems, automation, remote work, and digital project management tools.
Education:	Depended on textbooks, chalkboards, printed handouts, and classroom lectures.	Uses e-learning platforms, online classes, digital textbooks, and interactive learning tools.
Business Transactions:	Conducted in person, via postal mail, or through physical paperwork.	Conducted through e-commerce, online banking, digital payments, and virtual marketplaces.













DIGITAL TECHNOLOGIES – WHAT ARE THEY FOR?

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https://www.youtube.com/watch?v=CrL8lqJkvAE



FOR THE NATION

THE ROLE OF DIGITAL TECHNOLOGY

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The role of digital technology in various aspects of society:

- Global Connectivity and Communication
- Social Media and Online Communities
- Information Access and Knowledge Sharing
- Online Activism and Advocacy
- E-Governance and Citizen Engagement
- Virtual Reality and Augmented Reality



DIGITAL TECHNOLOGY IN EDUCATION

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Here are some examples of technology used in education:

- Video Conferencing
- Online Assessment and Grading Tools
- Gamification
- Artificial Intelligence (AI) in Education
- Online Simulations and Virtual Labs

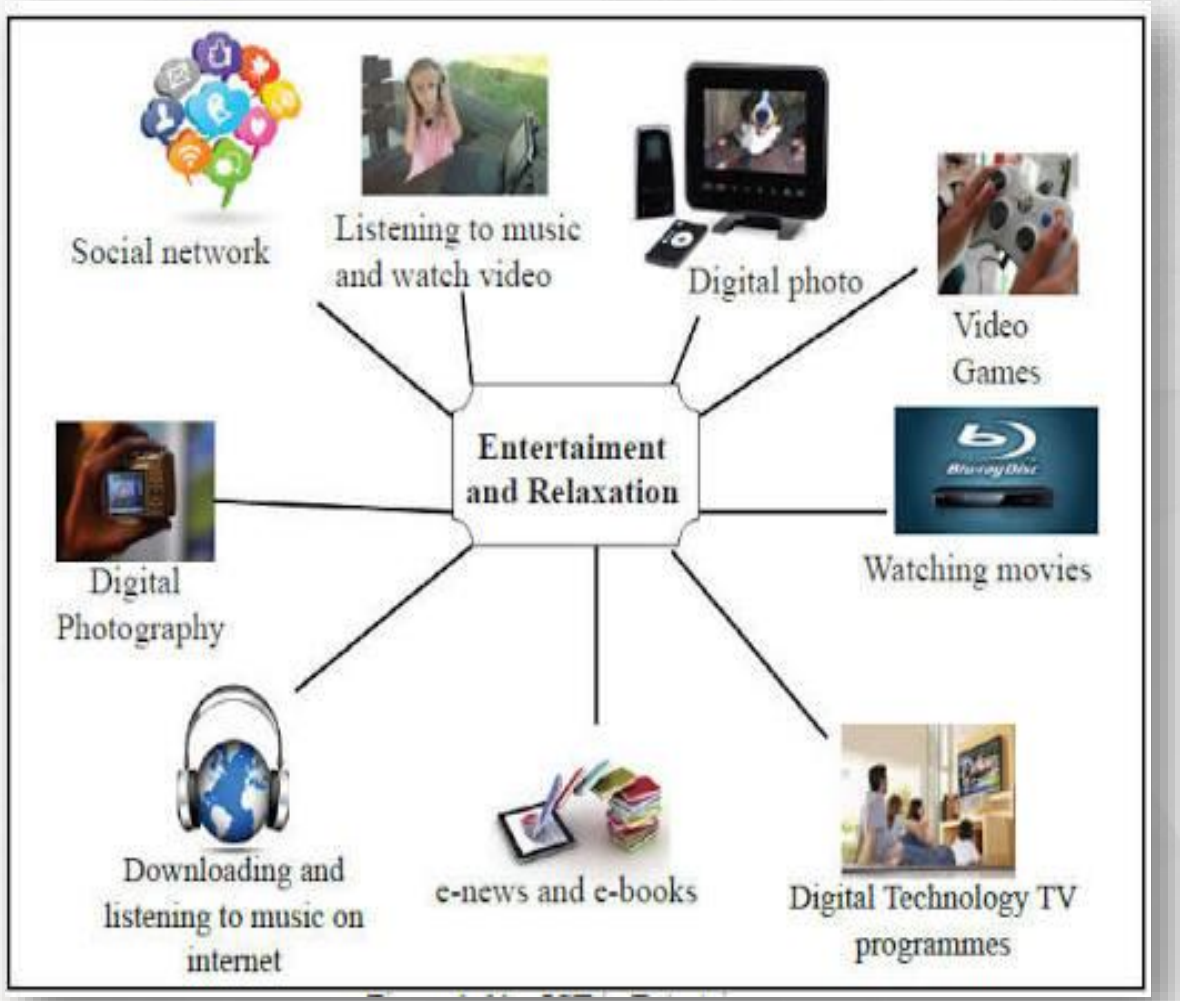


DIGITAL TECHNOLOGY IN ENTERTAINMENT /RECREATION

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Here are some examples of digital technology in entertainment and recreation:

- Streaming Services
- Online Gaming
- Digital Music Platforms
- Podcasts and Audiobooks
- Digital Comics and E-books
- Social Media Entertainment
- Live Streaming and Webcasts
- Digital Art and Creativity Tools



DIGITAL TECHNOLOGY IN TRAVEL AND TOURISM

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Here are some examples of digital technology in travel and tourism.

- Online Travel Booking Platforms
- Mobile Travel Apps
- Travel Review Websites
- Virtual Tours and 360-Degree Videos
- Augmented Reality (AR) Navigation
- Language Translation Apps

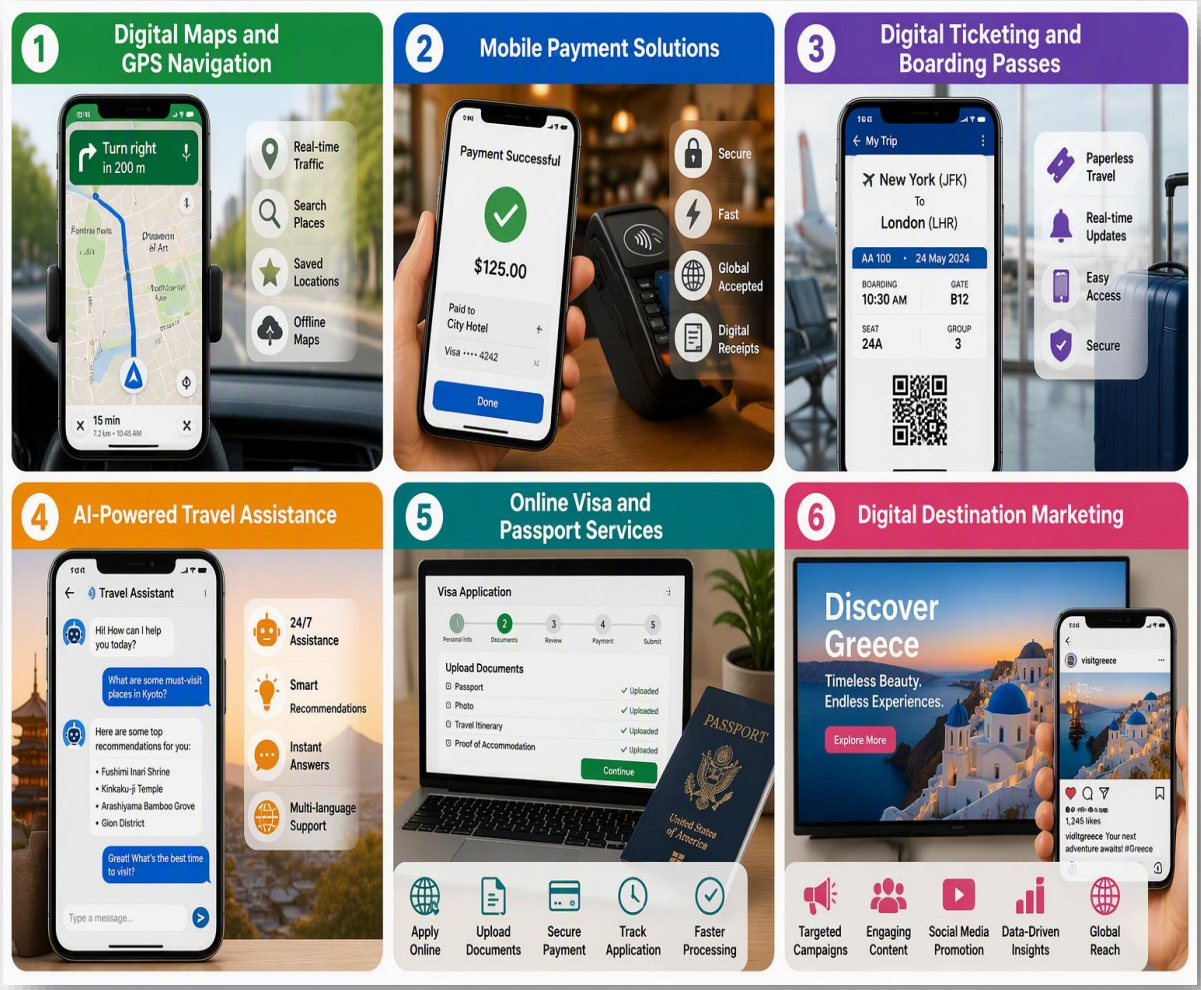


DIGITAL TECHNOLOGY IN TRAVEL AND TOURISM

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Here are some examples of digital technology in travel and tourism.

- Digital Maps and GPS Navigation
- Mobile Payment Solutions
- Digital Ticketing and Boarding Passes
- AI-Powered Travel Assistance
- Online Visa and Passport Services
- Digital Destination Marketing

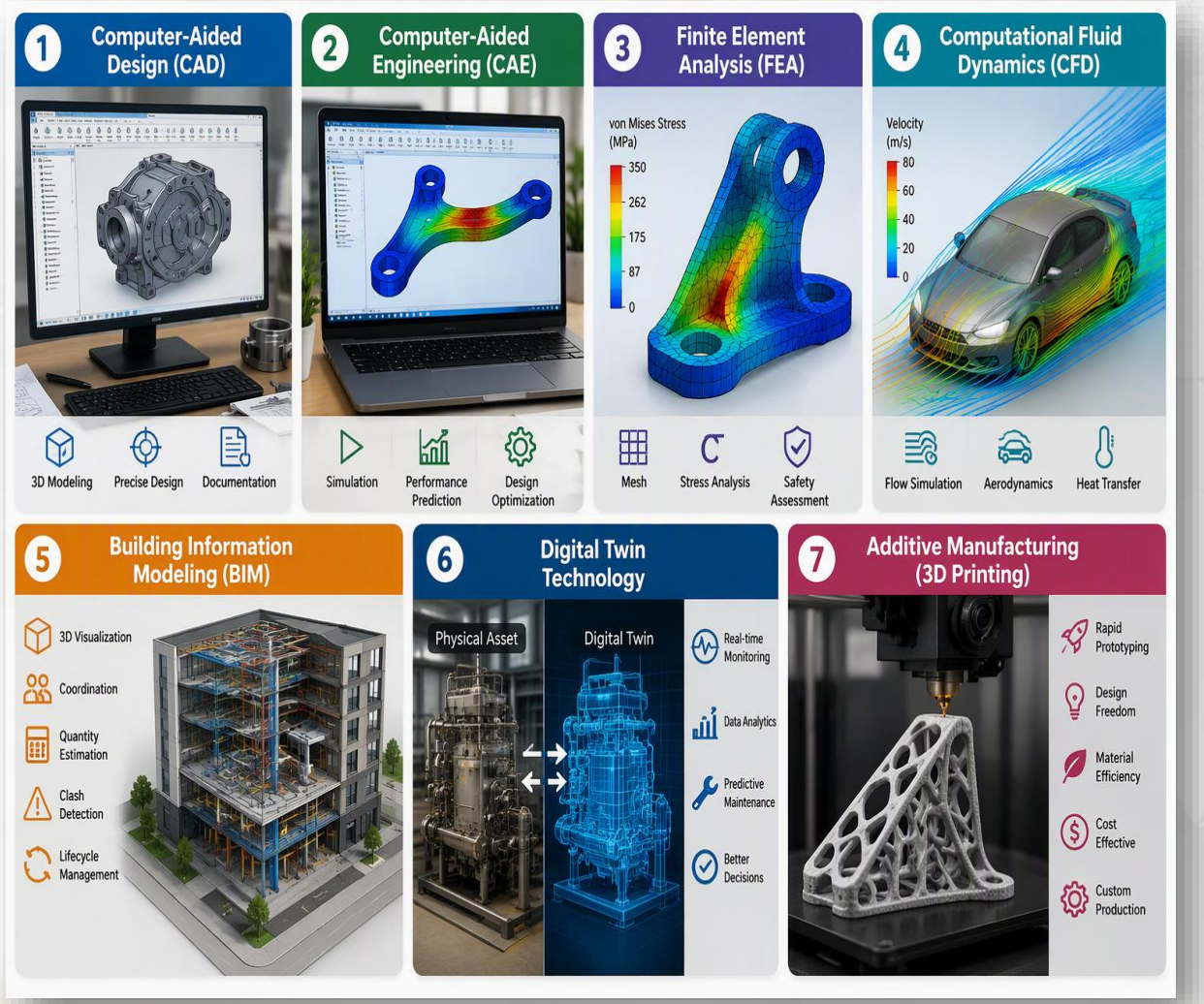


DIGITAL TECHNOLOGY IN ENGINEERING

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Here are some examples of digital technology in engineering:

- Computer-Aided Design (CAD)
- Computer-Aided Engineering (CAE)
- Finite Element Analysis (FEA)
- Computational Fluid Dynamics (CFD)
- Building Information Modeling (BIM)
- Digital Twin Technology
- Additive Manufacturing (3D Printing)



DIGITAL TECHNOLOGY IN ENGINEERING

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Here are some examples of digital technology in engineering:

- Internet of Things (IoT) in Engineering
- Robotics and Automation
- Geographical Information Systems (GIS)
- Energy Simulation and Optimization
- Project Management Software
- Remote Sensing and Surveying Technology
- Artificial Intelligence (AI) in Engineering
- Virtual Reality (VR) and Augmented Reality (AR) in Engineering



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DIGITAL TECHNOLOGY IN FINANCE

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Here are some examples of digital technology in Finance (FinTech):

- Online Banking
- Mobile Banking Apps
- Digital Payment Systems
- E-wallets
- Artificial Intelligence (AI)
- Blockchain Technology



DIGITAL TECHNOLOGY IN FINANCE

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Here are some examples of digital technology in Finance (FinTech):

- Cryptocurrency
- Cloud Computing
- Biometric Authentication
- Digital Lending Platforms
- InsurTech
- Big Data Analytics

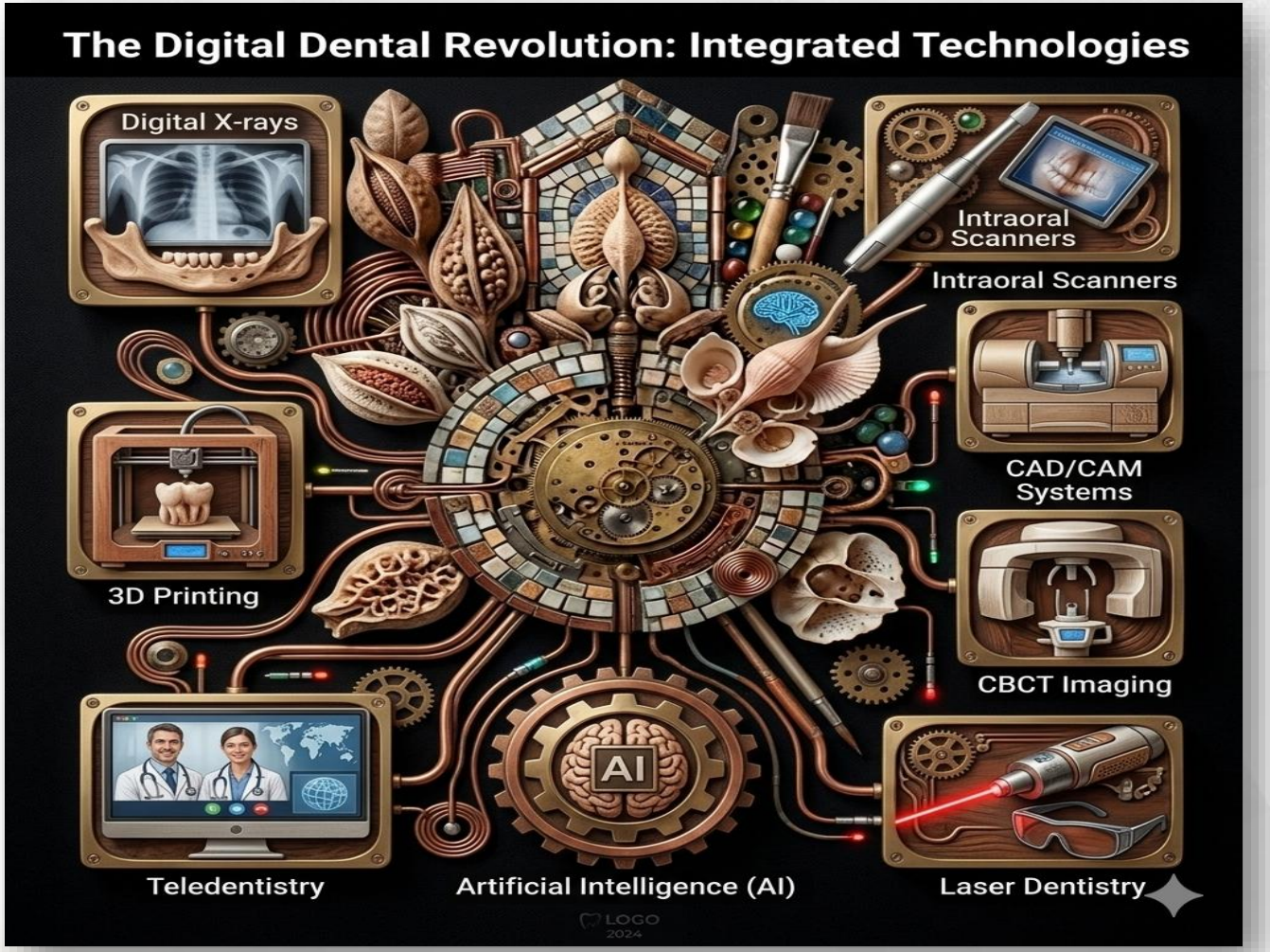


DIGITAL TECHNOLOGY IN DENTISTRY

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Here are some examples of digital technology in Dentistry:

- Digital X-rays
- Intraoral Scanners
- CAD/CAM Systems
- 3D Printing
- CBCT Imaging
- Laser Dentistry
- Artificial Intelligence (AI)
- Teledentistry

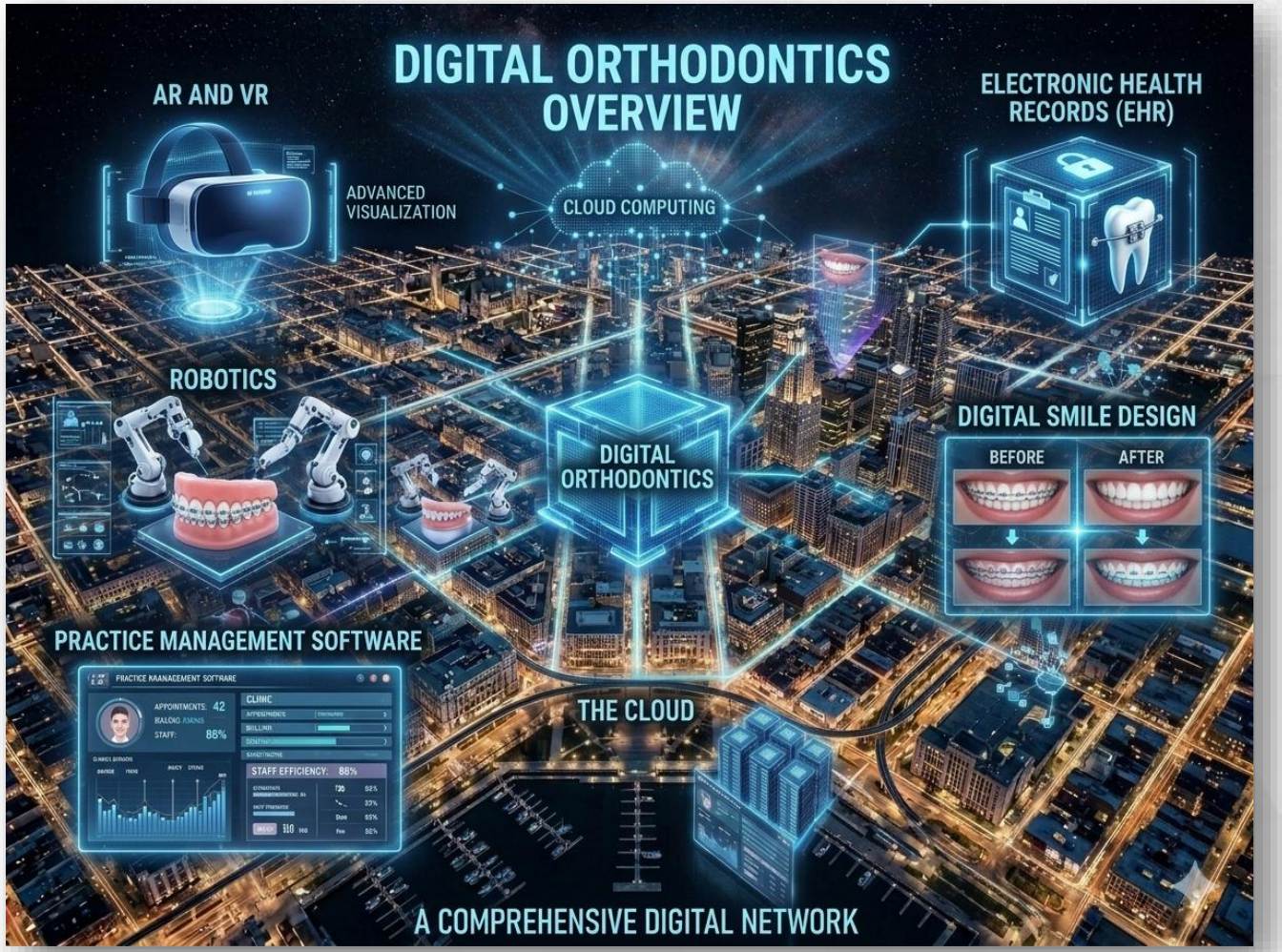


DIGITAL TECHNOLOGY IN DENTISTRY

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Here are some examples of digital technology in Dentistry:

- Digital Orthodontics
- Electronic Health Records (EHR)
- AR and VR
- Practice Management Software
- Robotics
- Cloud Computing
- Digital Smile Design



THREATS IN THE DIGITAL AGE

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Privacy Concerns



Personal information can be collected, shared, or misused without our consent.

Misinformation



False or misleading information can spread quickly and influence our beliefs and decisions.

Cyberbullying



Harmful, aggressive, or cruel behavior online can hurt others emotionally and mentally.

Digital Divide



Not everyone has equal access to technology and the internet, creating inequality in education, opportunities, and resources.

Digital Addiction



Excessive use of digital devices can lead to poor health, stress, sleep problems, and decreased productivity.

HOW TO HARNESS DIGITAL TECHNOLOGY

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Digital Literacy Education

Empowering people with the skills and knowledge to navigate the digital world safely and effectively.



- Learn how to find, evaluate and use information.
- Understand online safety, privacy, and digital ethics.
- Build skills for study, work, and communication online.

Data Privacy and Security Measures

Protecting personal information and using secure practices to stay safe online.



- Use strong passwords and change them regularly.
- Be careful about what information you share.
- Enable privacy settings and two-factor authentication.
- Keep your devices and software updated.

Combatting Misinformation and Fake News

Promoting critical thinking and responsible sharing to stop the spread of false information.



- Verify before you believe.
- Don't share unverified content.
- Report false information.
- Encourage others to think critically.

Internet Access for All

Ensuring everyone has affordable and equal access to the internet and digital opportunities.



- Expand internet infrastructure in all communities.
- Make internet services affordable and accessible.
- Provide devices and digital resources for those in need.
- Promote inclusion and equal opportunities online.

Responsible Use of Social Media

Using social media mindfully and respectfully to build a positive and safe online community.



- Be respectful and kind in your posts and comments.
- Protect your privacy and the privacy of others.
- Think before you post.
- Report and block harmful or inappropriate content.

HOW TO HARNESS DIGITAL TECHNOLOGY

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Online Safety and Cybersecurity Education

Empowering individuals with the knowledge and skills to stay safe and secure online.



- Use strong passwords and enable 2FA.
- Be aware of phishing and online scams.
- Protect personal information and privacy.
- Report suspicious activities.

Ethical Use of AI and Automation

Promoting responsible, fair, and human-centered use of AI and automation.



- Ensure fairness, transparency, and accountability.
- Avoid bias and respect human rights.
- Use AI to augment human potential, not replace it.
- Think ethically. Consider the impact.

Regulating Emerging Technologies

Establishing policies and frameworks to ensure safety, security, and societal benefit.



- Create clear laws and standards.
- Ensure safety, security, and privacy.
- Encourage innovation responsibly.
- Involve diverse stakeholders in decision-making.

Digital Inclusion and Accessibility

Ensuring everyone can access and benefit from digital technologies.



- Provide affordable internet and devices.
- Design accessible and inclusive technologies.
- Promote digital skills for all.
- Leave no one behind.

Environmental Sustainability in Technology

Minimizing the environmental impact of technology and promoting responsible use of resources.



- Reduce energy consumption and e-waste.
- Adopt sustainable practices in production and use.
- Choose long-lasting, repairable, and recyclable products.
- Support companies with eco-friendly initiatives.

Balancing Work-Life Integration

Using technology to create balance, well-being, and meaningful connections.



- Set boundaries between work and personal time.
- Avoid burnout. Take breaks and rest.
- Focus on what matters. Be present.
- Use technology to enhance, not overwhelm, life.

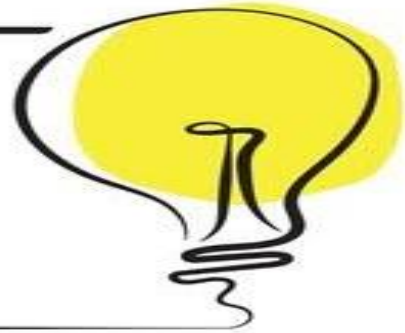
Encourage Green Technology

Innovating and adopting technologies that protect the planet and build a better future.



- Invest in clean energy and renewable solutions.
- Design efficient, low-impact technologies.
- Support green innovation and circular economy.
- Build a sustainable future for generations to come.

Conclusion



Society can fully capitalize on the potential of digital technology to drive positive change, innovation, and progress. Responsible and mindful use of digital technology can lead to a more inclusive, informed, and connected world, where technology truly serves as a force for societal benefit and well-being. As individuals and communities, we have the power to shape the digital age into a force that enriches lives, empowers individuals, and uplifts society.

Assignment

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Photography + Caption Challenge

- **Capture one photo** that symbolizes both a **positive** and a **negative** impact of digital technology.
- Example: A family sitting together but all on their phones (*connection vs. disconnection*).
- Add short captions explaining the message.

EXAMPLE:

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Thank you!

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